Sebastian De Leon

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Summary

Computer engineer with hands-on experience in full-stack development, embedded systems, and game development, focused on building reliable software and hardware-integrated solutions that solve real-world problems. Strong foundation in software engineering, rapid prototyping, with experience debugging complex issues and delivering scalable products. Passionate about intelligent systems, high-impact engineering and sustainable energy that advances performance, safety, and innovation.

Education

Western Michigan University Sep 2025 – Dec 2027
B.S. in Computer Engineering, GPA: 3.89 Kalamazoo, MI

Instituto Tecnológico de Santo Domingo Nov 2022 – Jul 2025
B.E. in Computer Science (transferred), GPA: 3.89 Santo Domingo, Dom. Rep.

Experience

Junior Software Engineer Jul 2024 – Jul 2026
Reispsolutions S.R.L. Santo Domingo, Dom. Rep.

- Architected a distributed Learning Management System (LMS) for a multi-brand restaurant corporation in the Dominican Republic, scaling training infrastructure across dozens of branch locations using ASP.NET.
- Built production-oriented ASP.NET Core services with .NET 9, CQRS, DDD, and Clean Architecture, implementing controllers, application services, domain models, and EF Core persistence across multiple backend services.
- Developed an end-to-end clinical management web application that automated patient intake flows, integrated calendar scheduling, and deployed automated SMS/email reminders for upcoming appointments.

Manufacturing & Telemetry Assistant Aug 2024 – Apr 2025
Instituto Tecnológico de Santo Domingo's NASA HERC College Team Santo Domingo, Dom. Rep.

- Designed and implemented Task A for the 2025 NASA HERC rover: a lead-screw actuator driven by a stepper motor that moved a sensor vertically to detect whether a container holding liquid contained water.
- Built the rover control system around an ESP32-S3, reading PWM signals from a commercial RC receiver and implementing a state machine to map controller inputs to rover motion and task execution.
- Led the full-stack architecture for the official Apolo 27 website, including frontend and backend development.
- Managed the development of proprietary games and digital tools for STEM outreach across the Dominican Republic.

Projects

Velo: Movie Matchmaker Sep 2025 – Present
Shipped mobile app React Native, Supabase

- Built a cross-platform Expo/React Native movie matchmaking app where groups create or join sessions, swipe on shared movie decks, and receive ranked watchlists using weighted voting, veto logic, and third-party API metadata.
- Designed a Clean Architecture codebase with TypeScript, Zustand, Supabase Auth/Postgres/Realtime/Storage, repository abstractions, dependency injection, and Edge Functions that securely proxy and cache third-party API data.
- Implemented production-focused features including solo discovery, QR/code-based session joining, persistent swipe progress, guest-to-account data migration, Letterboxd imports, taste-profile recommendations, five-language localization, and RevenueCat/AdMob monetization.

Printune May 2025 – Aug 2025
Tool to generate Spotify keychain 3D models Python, React, TypeScript

- Built a full-stack Next.js/TypeScript application that converts Spotify track, album, artist, and playlist URLs into personalized, 3D-printable keychains.
- Implemented browser-side image processing to extract Spotify Code bar dimensions, then used Three.js and React Three Fiber to generate interactive, orbitable 3D previews with customizable shapes, colors, and extrusion height.
- Developed STL export functionality supporting built-in base models and user-uploaded STL/GLB/GLTF files, enabling users to download print-ready keychains directly from the web app.

Marine Guardians May 2025 – Aug 2025
3D Interactive Simulation C#, Unity

- Architected scalable gameplay systems in Unity/C#, including a first-person controller, underwater movement, oxygen management, dynamic tool system, and game/player state handling using Unity's Input System.
- Built a data-driven wildlife discovery system using Unity ScriptableObjects, dynamically populating an in-game field guide with animal entries, scanned/unscanned states, icons, names, and educational descriptions without manually configuring each UI entry.

Technical Skills

- **Programming Languages**: C, C++, SQL, JavaScript, TypeScript
- **Frameworks & Tools**: Git, React, React Native, .NET, Linux, ESP-IDF, Docker
- **Software & Tools**: OrCAD X, KiCad, LTspice, SolidWorks, Fusion 360
- **Languages**: Spanish (native), English (Fluent)

Achievements & Certifications

- Harvard's CS50x: CS50's Introduction to Computer Science (edX)
- WMU College of Engineering and Applied Sciences Dean's List (all semesters)